OS-LAB ASSIGNMENT-03

1. Write a shell script to display current date in a particular format, number of users currently login and current month's calendar.

```
sachingola@DESKTOP-S4K7IL
sachingola@DESKTOP-S4K7IU0:~$ mkdir pakoda
sachingola@DESKTOP-S4K7IU0:~$ cd pakoda
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file1.sh
```

```
sachingola@DESKTOP-S4K7IL
 GNU nano 6.2                      file1.sh *
date +%T
date +%D
w|wc -l
cal
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file1.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file1.sh
19:29:51
02/19/24
2
   February 2024
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29
```

2. Write a shell script to display the process name and its process id.

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file2.sh
```

```
sachingola@DESKTOP-S4K7IL
 GNU nano 6.2                      file2.sh
echo
echo List of all processes with their ID and names:
echo
ps -eo pid,comm
echo
echo

echo Process ID of the current Shell:
echo $$
echo

#Note:We are using 'echo' to print new blank lines.
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file2.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file2.sh

List of all processes with their ID and names:

  PID COMMAND
    1 init
   13 init
   14 init
   15 bash
   91 bash
   92 ps


Process ID of the current Shell:
91
```

3. Write a shell script to take name as a input and display a greeting message to the user by checking system clock.

```
sachingola@DESKTOP-S4K7IL
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file3.sh
```

```
sachingola@DESKTOP-S4K7IL
  GNU nano 6.2                          file3.sh
echo

echo Enter your name:
read name
current_time=$(date +%H)

if [ $current_time -lt 12 ]; then
echo "Hello $name Good Morning."

elif [ $current_time -ge 18 ]; then
echo "Hello $name Good Evening."

else echo "Hello $name Good Afternoon."

fi

echo


#Note: The single 'echo' command is used to print blank line.
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file3.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file3.sh

Enter your name:
Sachin Gola
Hello Sachin Gola Good Evening.
```

4. Write a shell script to merge the content of 2 files into one file.

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ cat > first.txt
If nothing goes right , go left.
^C
sachingola@DESKTOP-S4K7IU0:~/pakoda$ cat > second.txt
If you are in hell, good luck and continue to be in.
^C
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ touch merged_file.txt
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file4.sh
```

```
sachingola@DESKTOP-S4K7IL
  GNU nano 6.2                                  file4.sh
# Receiving names of files( must exist) from user in three variables :
echo
read file1
read file2
read merged_file

# Merging the content of the files into one:

cat "$file1" "$file2" > "$merged_file"
echo

echo "The contents of  given files have been merged. The content of merged file is merged file."
echo

cat "$merged_file"
echo
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file4.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file4.sh

first.txt
second.txt
merged_file.txt

The contents of  given files have been merged. The content of merged file is merged file.

If nothing goes right , go left.
If you are in hell, good luck and continue to be in.
```

5. Write a shell script to create a tsv file containing name, roll no. and age of 10 students. Then use that tsv file to display only the names of the students in alphabetical order.

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file5.sh
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ touch new_file.tsv
```

```
sachingola@DESKTOP-S4K7IL
  GNU nano 6.2                                  file5.sh *
#Given file must exist
filename="new_file.tsv"

echo -e "Name\tRoll_No\tAge" > "$filename"
echo

#Using For loop  to get data from user:

for ((i=1; i<=10; i++)); do
read -p "Name: " data1
read -p "Roll number: " data2
read -p "Age :" data3

# Appending data in file after each iteration:
echo -e "$data1\t$data2\t$data3" >> "$filename"
echo
done


#printing  names in alphabetical order:

echo "Names of students in alphabetical order:"

cut -f 1 "$filename" | sort
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file5.sh

Name: Aman
Roll number: 1
Age :19

Name: Bhawna
Roll number: 2
Age :20

Name: Chetan
Roll number: 3
Age :18

Name: Farhan
Roll number: 4
Age :21

Name: Gauri
Roll number: 6
Age :19

Name: Himani
Roll number: 7
Age :19

Name: Jagdish
Roll number: 8
Age :20

Name: Mayank
Roll number: 9
Age :18

Name: Payal
Roll number: 10
Age :20

Name: Raunak
Roll number: 11
Age :19

 Names of students in alphabetical order:
Aman
Bhawna
Chetan
Farhan
Gauri
Himani
Jagdish
Mayank
Name
Payal
Raunak
```

6. You are given a file of tab-delimited weather data (TSV). There is no header column in this data file. The first five columns of this data are: (a) the name of the city (b) the average monthly temperature in Jan (in Fahrenheit). (c) The average monthly temperature in April (in Fahrenheit). (d) The average monthly temperature in July (in Fahrenheit). (e) the average monthly temperature in October (in Fahrenheit).You need to sort this file on the basis of average monthly temperature in April.

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ touch sixth_file.tsv
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file6.sh
```

sachingola@DESKTOP-S4K7IL

```
  GNU nano 6.2                          file6.sh
#Given file must exist
filename="sixth_file.tsv"

echo

#Using For loop  to get data from user:

for ((i=1; i<=5; i++)); do
read -p "Name of the city: " data1
read -p "Avg. temp. of January(in F): " data2
read -p "Avg. temp. of April(in F): " data3
read -p "Avg. temp. of July(in F): " data4
read -p "Avg. temp. of October(in F): " data5


# Appending data in file after each iteration:
echo -e "$data1\t$data2\t$data3\t$data4\t$data5" >> "$filename"
echo
done



echo "Sorting file on the basis of average temperature in April: "
echo

sort -k3,3 -n "$filename"
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file6.sh
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file6.sh

Name of the city: Agra
Avg. temp. of January(in F): 40
Avg. temp. of April(in F): 85
Avg. temp. of July(in F): 96
Avg. temp. of October(in F): 48

Name of the city: Delhi
Avg. temp. of January(in F): 35
Avg. temp. of April(in F): 90
Avg. temp. of July(in F): 100
Avg. temp. of October(in F): 58

Name of the city: Sydney
Avg. temp. of January(in F): 78
Avg. temp. of April(in F): 85
Avg. temp. of July(in F): 46
Avg. temp. of October(in F): 87

Name of the city: Dubai
Avg. temp. of January(in F): 78
Avg. temp. of April(in F): 96
Avg. temp. of July(in F): 89
Avg. temp. of October(in F): 85

Name of the city: Ambala
Avg. temp. of January(in F): 30
Avg. temp. of April(in F): 50
Avg. temp. of July(in F): 74
Avg. temp. of October(in F): 64

Sorting file on the basis of average temperature in April:

Ambala  30      50      74      64
Agra    40      85      96      48
Sydney  78      85      46      87
Delhi   35      90      100     58
Dubai   78      96      89      85
```